University of Utah Spring 2026

MATH 2270-006
Midterm 2 Questions

Instructor: Alp Uzman

February 27 2026, 9:40 AM - 10:30 AM

Surname: KEY

First Name:

uNID:

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1. **[25 points]** Consider the following set of four dimensional vectors:

$$S = \left\{ \overbrace{\begin{pmatrix} 1 \\ -1 \\ -2 \\ 5 \end{pmatrix}}^{a_1}, \overbrace{\begin{pmatrix} -1 \\ 4 \\ -7 \\ 7 \end{pmatrix}}^{a_2}, \overbrace{\begin{pmatrix} 0 \\ 2 \\ -6 \\ 8 \end{pmatrix}}^{a_3}, \overbrace{\begin{pmatrix} 2 \\ -3 \\ -1 \\ 6 \end{pmatrix}}^{a_4}, \overbrace{\begin{pmatrix} 3 \\ -8 \\ 9 \\ -5 \end{pmatrix}}^{a_5} \right\}$$

(a) **[15 points]** Find a basis for Span(S). $= \mathrm{Col}(A)$ if $A = (a_1\ a_2 \cdots a_5)$

$$\begin{pmatrix} 1 & -1 & 0 & 2 & 3 \\ -1 & 4 & 2 & -3 & -8 \\ -2 & -7 & -6 & -1 & 9 \\ 5 & 7 & 8 & 6 & -5 \end{pmatrix} \to \begin{pmatrix} 1 & -1 & 0 & 2 & 3 \\ 0 & 3 & 2 & -1 & -5 \\ 0 & -9 & -6 & 3 & 15 \\ 0 & 12 & 8 & -4 & -20 \end{pmatrix}$$

$$\downarrow$$

$$\begin{pmatrix} 1 & -1 & 0 & 2 & 3 \\ 0 & 3 & 2 & -1 & -5 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \leftarrow \begin{pmatrix} 1 & -1 & 0 & 2 & 3 \\ 0 & 3 & 2 & -1 & -5 \\ 0 & +3 & 2 & -1 & -5 \\ 0 & 3 & 2 & -1 & -5 \end{pmatrix}$$

pivots

$$\to \boxed{\left\{ \begin{pmatrix} 1 \\ -1 \\ -2 \\ 5 \end{pmatrix}, \begin{pmatrix} -1 \\ 4 \\ -7 \\ 7 \end{pmatrix} \right\} \text{ basis for Span(S)}}$$

(b) **[10 points]** Determine if S is linearly independent.

$\boxed{\text{No}}$: $\#(S) = 5 > \dim(\mathbb{R}^4) = 4$.

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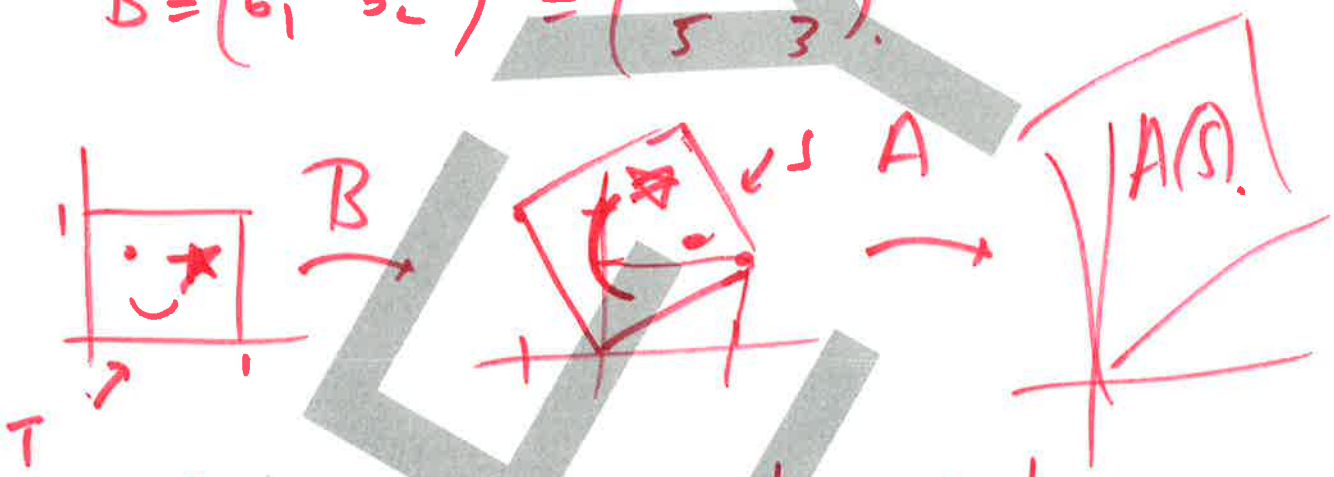
2. **[10 points]** Let S be the parallelogram whose corners are

$$\begin{pmatrix}0\\0\end{pmatrix}, \quad b_1 = \begin{pmatrix}-3\\5\end{pmatrix}, \quad b_2 = \begin{pmatrix}-4\\3\end{pmatrix}, \quad b_1 + b_2,$$

and let $A = \begin{pmatrix}3 & 2\\-4 & 6\end{pmatrix}$. Compute the area of the image of S under the linear transformation A.

$B = (b_1 \ b_2) = \begin{pmatrix}-3 & -4\\5 & 3\end{pmatrix}.$



$\text{area}(A(S)) = \text{area}(S) \cdot |\det(A)|$

$= \text{area}(T) \cdot |\det(B)|\,|\det(A)|$

$1 \quad |-9+20| \quad 18+8$

$= 11 \cdot 26 = 286.$

$$\begin{array}{r} 26 \\ 11 \\ \hline 26 \\ 26 \\ \hline 286 \end{array}$$

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3. **[20 points]** Compute the determinant of the following matrix:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 4 & 6 \end{pmatrix}.$$

$$\det(A) = (\cancel{\ast}) - (\cancel{\ast})$$

$$= 1\cdot 3\cdot 6 + 2\cdot 5\cdot 3 + \underline{2\cdot 4\cdot 3}$$

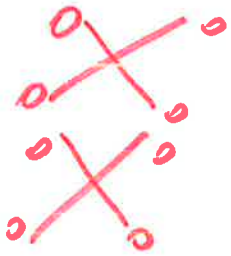
$$- 3\cdot 3\cdot 3 - \underline{2\cdot 2\cdot 6} - 1\cdot 4\cdot 5$$

$$= 18 - 20 + 10\cdot 3 - 9\cdot 3$$

$$= -2 + 3 = 1$$

4. **[15 points]** Consider the permutation P defined by

$$P\begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} = \begin{pmatrix} y \\ x \\ t \\ z \end{pmatrix}.$$



(a) **[5 points]** Write the permutation P in matrix form.

$$P = \left(\begin{array}{cc|cc} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{array}\right)$$

(b) **[5 points]** Compute the determinant of P.

$$\det(P) = (-1)(-1) = 1$$

(c) **[5 points]** Compute the inverse of P.

$$P^{-1} = P^{T} = P.$$

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5. **[30 points]** Consider the following matrix:

$$A = \begin{pmatrix} 1 & -2 & 5 & 4 \\ 1 & -1 & 5 & -3 \\ -2 & 0 & 1 & -2 \\ 4 & 1 & 1 & -9 \end{pmatrix}$$

$$\begin{array}{r} 9 \\ \times\ 7 \\ \hline 63 \end{array} \qquad \begin{array}{r} 63 \\ -25 \\ \hline 38 \end{array}$$

Perform the following tasks:

(a) **[10 points]** Compute the nullity of A.

$$\begin{pmatrix} 1 & -2 & 5 & 4 \\ 1 & -1 & 5 & -3 \\ -2 & 0 & 1 & -2 \\ 4 & 1 & 1 & -9 \end{pmatrix} \xrightarrow{E_1} \begin{pmatrix} 1 & -2 & 5 & 4 \\ 0 & 1 & 0 & -7 \\ 0 & -4 & 11 & 6 \\ 0 & 9 & -19 & -25 \end{pmatrix} \xrightarrow{E_2} \begin{pmatrix} 1 & 0 & 5 & -10 \\ 0 & 1 & 0 & -7 \\ 0 & 0 & 11 & -22 \\ 0 & 0 & -19 & +38 \end{pmatrix}$$

$$\xrightarrow{E_3} \begin{pmatrix} 1 & 0 & 5 & -10 \\ 0 & 1 & 0 & -7 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 1 & -2 \end{pmatrix} \xrightarrow{E_4} \begin{pmatrix} 1 & 0 & 5 & -10 \\ 0 & 1 & 0 & -7 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$\Rightarrow$ $\boxed{\text{nullity} = 1}$

(b) **[10 points]** Compute the determinant of A.

$\text{rank}(A) = 4 - 1 = 3 < 4$

$\Rightarrow$ A not inv.

$\Rightarrow$ $\boxed{\det(A) = 0}$

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(c) **[6 points]** Find two matrices L and U satisfying the following properties simultaneously:

- A = LU,
- L is invertible, and
- U is an echelon form of A.

$$U = \begin{pmatrix} 1 & 0 & 5 & -10 \\ 0 & 1 & 0 & -7 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

echelon.

From part (a),

$$E_4 E_3 E_2 E_1 A = U.$$

$$\Rightarrow A = E_1^{-1} E_2^{-1} E_3^{-1} E_4^{-1} U$$

$$E_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 2 & 0 & 1 & 0 \\ -4 & 0 & 0 & 1 \end{pmatrix}, \quad E_2 = \begin{pmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 4 & 1 & 0 \\ 0 & -9 & 0 & 1 \end{pmatrix}, \quad E_3 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1/11 & 0 \\ 0 & 0 & 0 & -1/19 \end{pmatrix}, \quad E_4 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -11 \end{pmatrix}$$

$$E_1^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ -2 & 0 & 1 & 0 \\ 4 & 0 & 0 & 1 \end{pmatrix}, \quad E_2^{-1} = \begin{pmatrix} 1 & -2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -4 & 1 & 0 \\ 0 & 9 & 0 & 1 \end{pmatrix}, \quad E_3^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 11 & 0 \\ 0 & 0 & 0 & -19 \end{pmatrix}, \quad E_4^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

$$L = E_1^{-1} E_2^{-1} E_3^{-1} E_4^{-1} = \begin{pmatrix} 1 & -2 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ -2 & 0 & 1 & 0 \\ 4 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 11 & 0 \\ 0 & 0 & -19 & -19 \end{pmatrix} = \underline{\begin{pmatrix} 1 & -2 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ -2 & 0 & 11 & 0 \\ 4 & 1 & -19 & -19 \end{pmatrix}}$$

L inv.

(d) **[4 points]** Find three matrices L, D, and R satisfying the following properties simultaneously:

- A = LDR,
- L is invertible,
- D is diagonal, and
- R is invertible.

$$L = \begin{pmatrix} 1 & -2 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ -2 & 0 & 11 & 0 \\ 4 & 1 & -19 & -19 \end{pmatrix}$$

$$U = \begin{pmatrix} 1 & 0 & 5 & -10 \\ 0 & 1 & 0 & -7 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

← from ℓ partia.

$$U \underbrace{\begin{pmatrix} 1 & 0 & -5 & 10 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}}_{F_1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -7 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$UF_1 \underbrace{\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}}_{F_2} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$UF_1F_2 \underbrace{\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix}}_{F_3} = \underbrace{\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}}_{=D}$$

$$\Rightarrow U = D F_3^{-1} F_2^{-1} F_1^{-1} = D \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -7 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} F_1^{-1} = D \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -7 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 5 & -10 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

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$$\Rightarrow \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -7 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 5 & -10 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$= D \underbrace{\begin{pmatrix} 1 & 0 & 5 & -10 \\ 0 & 1 & 0 & -7 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{pmatrix}}_{R \text{ inv.}}$$

$$\Rightarrow A = LU = \boxed{LDR}$$

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